NAME: _____ Geometry (FL B.E.S.T.)

9.5 Arc Length and Sector Area

A central angle θ claims the fraction $\dfrac{\theta}{360^\circ}$ of an entire circle. Apply that fraction to the circle's circumference to get **arc length**, $\ell = \dfrac{\theta}{360^\circ} \cdot 2\pi r$, or to the circle's area to get **sector area**, $A = \dfrac{\theta}{360^\circ} \cdot \pi r^2$. A useful shortcut follows directly: since both formulas share the same fraction, $A = \dfrac{1}{2} r\ell$ — a sector's area is half its radius times its arc length, just like a triangle's area is half its base times its height. This worksheet leaves answers in terms of π when the numbers work out exactly, and uses the arc–area shortcut to solve for two unknowns at once.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Radius 4, angle 90°. Find the arc length.

2. Radius 8, angle 45°. Find the sector area.

3. Radius 10, angle 180°. Find the arc length.

4. Radius 6, angle 60°. Find the sector area.

5. Radius 9, arc length 3π. Find θ.

6. Sector area 12π, angle 90°. Find r.

7. Radius 4, angle 150°. Find the arc length and sector area.

8. Radius 12, angle 30°. Find the arc length.

9. Radius 5, angle 144°. Find the sector area.

10. Radius 6, arc length 2π. Find θ.

11. Arc length 4π, angle 120°. Find r.

12. Radius 7, angle 180°. Find the sector area.

13. What fraction of a circle is a 40° sector?

14. A sector has arc length 6π and area 30π. Find the radius and the central angle.

15. A rotating sprinkler sprays water out to 15 ft and rotates through 150°. Find the area watered (leave in terms of π, then approximate using $\pi \approx 3.14$) and the arc length of the spray's edge (in terms of π).

✓ Answer Key

1. 2π
2. 8π
3. 10π
4. 6π
5. 60°
6. $4\sqrt{3}$
7. Arc $\frac{10\pi}{3}$; area $\frac{20\pi}{3}$
8. 2π
9. 10π
10. 60°
11. $r = 6$
12. $\frac{49\pi}{2}$
13. $\frac{1}{9}$
14. $r = 10$; $\theta = 108^\circ$
15. Area $= 93.75\pi \approx 294$ ft^2; arc length $= 12.5\pi$ ft